

Notes: Gale and Hellwig (RES, 1985)

Incentive-Compatible Debt Contracts: The One-Period Problem

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1 Big picture

- **Research question.** What form should an optimal credit contract take when the borrower observes the project outcome privately and the lender can verify the outcome only by paying a cost? Does this informational friction generate credit rationing through underinvestment?
- **Main idea.** The paper treats the firm's budget constraint as a contract-design problem. Instead of assuming debt and bankruptcy, it asks whether those arrangements arise endogenously from incentive compatibility and costly verification.
- **Core environment.** An entrepreneur needs outside finance to invest in a risky project. The entrepreneur observes the state of nature for free after investment; the investor can observe it only at a positive cost. Verification is interpreted as bankruptcy.
- **Main contract result.** Under the baseline assumptions, the optimal incentive-compatible contract is a standard debt contract: the borrower pays a fixed amount when solvent, the lender verifies only when the borrower cannot pay, and the lender recovers as much as possible in bankruptcy.
- **Main investment result.** The second-best investment level never exceeds the first-best level. If bankruptcy has a positive probability and verification/bankruptcy is costly at the margin, the second-best investment level is strictly below the first best.
- **Credit rationing.** The relevant rationing is not necessarily denial of credit. In this competitive-capital-market model, rationing appears as a smaller loan and lower investment than the first-best project would require.
- **Comparison with price taking.** A borrower who treats the interest rate as exogenous generally chooses a contract with a higher bankruptcy probability than the optimal contract. The investment comparison is ambiguous because a higher stated rate can both discipline borrowing and increase default risk.
- **Liquidity result.** More borrower net wealth need not monotonically raise investment. Under some conditions the relationship between net wealth and optimal investment can be nonmonotonic.

Economic takeaway. The paper gives a canonical costly-state-verification foundation for debt. Debt is optimal because it economizes on verification: the lender is paid a fixed amount in good states and verifies only in bad states, but this economizing on bankruptcy costs distorts investment downward.

2 Incremental contribution of the paper

- **Endogenizes debt.** Earlier credit-market work often assumed a debt contract. Gale and Hellwig derive the standard debt contract from a formal contracting problem with asymmetric information and costly observation.
- **Bankruptcy as information acquisition.** The paper gives bankruptcy a precise role: it is the costly act through which the lender discovers the true project outcome and seizes the recoverable part of revenue.

- **Separates contract shape from contract position.** The authors ask both what the optimal repayment schedule looks like and where it is located, meaning how large the investment and loan should be.
- **Credit rationing without indivisible projects.** In contrast to models in which rationing means denying some firms credit, this model has divisible investment. Rationing appears as underinvestment relative to the first best.
- **Risk-neutral inefficiency.** The baseline inefficiency does not rely on risk aversion. It arises because state verification is costly and the project outcome is privately observed by the borrower.
- **Comparison with interest-rate-taking behavior.** The paper distinguishes the optimal contract chosen in a competitive contract market from a hypothetical borrower who treats the loan rate as fixed and ignores how investment choices affect lender break-even.
- **Liquidity comparative statics.** The paper cautions against a simple monotone view that more internal wealth always raises investment. The efficient contract set can generate nonmonotonic investment responses.
- **Foundation for later corporate finance theory.** The paper is a key predecessor to the modern view that debt, default, and financial constraints can be derived from information and verification frictions.

3 Model and measurement

3.1 Agents, timing, and objective

The model has one representative entrepreneur and one investor in a competitive capital market. The entrepreneur owns a firm but lacks enough liquid resources to finance the efficient project scale. The investor is interpreted as a bank or other diversified financial institution.

- **Date 0.** The contract is written, investment I is chosen, and financing is arranged.
- **Date 1.** The state s is realized and project revenue $f(s, I)$ is produced.
- **Borrower information.** The entrepreneur observes s at no cost.
- **Lender information.** The investor can observe s only by paying an observation cost.
- **Baseline preferences.** Both agents are risk neutral in the main model, with the investor able to diversify perfectly across many loans.

The entrepreneur's initial net wealth is

$$W_0 = A_0 - R_0, \tag{1}$$

where A_0 is liquid assets and R_0 is initial debt. The investor can raise funds at the riskless opportunity cost i .

3.2 Technology and observation costs

Investment produces revenue $f(s, I)$. The paper assumes positive but diminishing marginal returns to investment:

$$D_2f(s, I) > 0, \quad D_{22}f(s, I) < 0, \quad D_{12}f(s, I) > 0. \quad (2)$$

The cross-partial condition means better states raise the marginal return to investment.

Observation has two costs:

- $c_0 \geq 0$: a fixed, nonpecuniary cost imposed on the entrepreneur when the lender observes the state.
- $c_1(s, I) \geq 0$: the lender's observation or bankruptcy cost, which may depend on the state and investment.

Interpretation. If observation is bankruptcy, then c_1 can capture the loss of going-concern value, liquidation costs, managerial disruption, legal costs, and damage to customer or employee relationships.

3.3 The contract

A contract is

$$(I, C_0, C_1, W_1, B), \quad (3)$$

where:

- I is investment.
- C_0 is the entrepreneur's date-0 contribution, interpreted as equity.
- $L = I + R_0 - C_0$ is the new loan supplied by the investor.
- $C_1(s)$ is the investor's date-1 return.
- $W_1(s)$ is the entrepreneur's date-1 wealth.
- $B(s)$ equals one if the investor observes/verifies the state and zero otherwise.

The first-best investment level is the complete-information benchmark:

$$I^* \in \arg \max_{I \geq 0} E[f(s, I) - (1 + i)I]. \quad (4)$$

Assumption 3 rules out the trivial case in which the entrepreneur can self-finance the first-best project.

3.4 Incentive compatibility

Because the entrepreneur privately observes s , the contract must give him no incentive to misreport. The key incentive-compatible implication is that the lender's return must be constant in all nonverified states:

$$B(s) = 0 \quad \Rightarrow \quad C_1(s) = R_1. \quad (5)$$

Intuition. If two nonverified states promised different payments to the lender, the entrepreneur would announce the lower-payment state whenever possible. Therefore, no-verification states must look identical from the lender's payoff perspective.

When the true state is a verified state but the entrepreneur could instead announce a nonverified state, the contract must make such misreporting unattractive. This is why the verified-state transfer must absorb the verification cost at the margin.

3.5 Maximum equity participation

A contract has maximum equity participation (MEP) if $C_0 = A_0$. The paper shows any optimal contract is weakly dominated by one with MEP.

Interpretation. In the formal model, the exact split between date-0 equity and date-1 repayment is partly indeterminate because liquid assets can be used to repay debt. For the bankruptcy interpretation, however, putting all liquid assets into the project helps reduce the promised repayment and the probability of costly bankruptcy.

4 Methodology and theoretical specifications

4.1 Optimal contract problem

The competitive capital market selects the contract that maximizes the entrepreneur's expected payoff subject to the investor's break-even condition and feasibility:

$$\max E[W_1 - c_0 B] \quad (6)$$

$$\text{s.t. } EC_1 \geq (1 + i)(I + R_0 - C_0), \quad (7)$$

$$C_1 + W_1 \leq f - c_1 B + (1 + i)(A_0 - C_0), \quad (8)$$

$$I \geq 0, \quad 0 \leq C_0 \leq A_0, \quad W_1 \geq 0, \quad (9)$$

$$(I, C_0, C_1, W_1, B) \text{ is incentive-compatible.} \quad (10)$$

After eliminating W_1 and using the investor's break-even constraint, the problem becomes an investment-and-repayment problem in which expected surplus subtracts the expected verification costs:

$$\max E[f - (1 + i)I - (c_0 + c_1)B] \quad (11)$$

$$\text{s.t. } EC_1 = (1 + i)(I + R_0 - C_0), \quad (12)$$

$$C_1 \leq f + (1 + i)(A_0 - C_0) - c_1 B, \quad (13)$$

$$I \geq 0, \quad 0 \leq C_0 \leq A_0, \quad (14)$$

$$(I, C_0, C_1, B) \text{ is incentive-compatible.} \quad (15)$$

4.2 Standard debt contract

The paper defines a standard debt contract (SDC) by three properties:

$$\text{Fixed repayment: } (1 - B)(C_1 - R_1) = 0, \quad (16)$$

$$\text{Bankruptcy decision: } B = 1 \iff f < R_1, \quad (17)$$

$$\text{Maximum recovery: } BC_1 = B(f - c_1). \quad (18)$$

Economic intuition. The contract avoids verification whenever the borrower can make the fixed repayment R_1 . Verification happens only when the reported ability to pay is insufficient. In bankruptcy, the lender receives all recoverable revenue because this allows the fixed repayment in solvent states to be as low as possible, reducing the probability of costly verification.

4.3 Why debt is optimal

The proof logic has three steps.

- **Step 1: Nonverified states must have a flat payment.** Incentive compatibility forces $C_1(s) = R_1$ whenever $B(s) = 0$.
- **Step 2: Verified states should maximize recovery.** If the lender recovers less than possible in bankruptcy, the same expected lender return requires a higher R_1 in nonbankruptcy states, which increases bankruptcy probability and expected verification costs.
- **Step 3: Verification should occur exactly in low-output states.** If the state is good enough to pay R_1 , verifying it wastes resources. If the state is too low to pay R_1 , verification is needed to discipline the borrower's report and recover assets.

Together these steps deliver the SDC with MEP. The converse requires positive observation costs around the repayment threshold so that unnecessary verification is strictly wasteful.

4.4 Bankruptcy point representation

With an SDC, the contract can be summarized by investment I and a bankruptcy cutoff γ . The promised repayment is

$$R_1 = f(\gamma, I), \quad (19)$$

and bankruptcy occurs when $s < \gamma$.

Define two payoff functions:

$$f_\gamma(s, I) = \begin{cases} f(s, I), & s \geq \gamma, \\ f(s, I) - c_1(s, I) - c_0, & s < \gamma, \end{cases} \quad (20)$$

$$g_\gamma(s, I) = \begin{cases} f(\gamma, I), & s \geq \gamma, \\ f(s, I) - c_1(s, I), & s < \gamma. \end{cases} \quad (21)$$

Then the problem reduces to:

$$\max_{I, \gamma} E[f_\gamma(s, I) - (1 + i)I] \quad (22)$$

$$\text{s.t. } Eg_\gamma(s, I) \geq (1 + i)(I - W_0), \quad (23)$$

$$I \geq 0, \quad \gamma \geq 0. \quad (24)$$

Interpretation. f_γ is total surplus net of bankruptcy costs. g_γ is the lender's recoverable cash flow. The constraint says the lender must expect to recover the opportunity cost of funds.

4.5 Underinvestment and credit rationing

For a fixed bankruptcy cutoff γ , define:

$$I^*(\gamma) = \arg \max_{I \geq 0} E[f_\gamma(s, I) - (1 + i)I], \quad (25)$$

$$\underline{I}(\gamma) = \arg \max_{I \geq 0} E[g_\gamma(s, I) - (1 + i)I]. \quad (26)$$

The optimal investment satisfies:

$$\underline{I}(\gamma) \leq I \leq I^*(\gamma) \leq I^*. \quad (27)$$

The strict underinvestment result requires both a positive probability of bankruptcy and costly bankruptcy. If bankruptcy occurs with positive probability and increasing investment raises bankruptcy costs at the margin, then

$$I < I^*. \quad (28)$$

Economic intuition. Extra investment increases output in all states, but in bankruptcy states part of output is lost to verification or bankruptcy costs. Therefore the private contract trades off productive investment against expected bankruptcy losses. Reducing investment below the first best can reduce the bankruptcy region and expected verification waste.

4.6 When first best can still occur

The first-best investment can still be achieved in special cases:

- Bankruptcy has no cost, so verification does not waste resources.
- The borrower has enough wealth or sufficiently high guaranteed output to finance first-best investment without a positive probability of bankruptcy.
- The relevant marginal bankruptcy costs are zero.

Interpretation. Asymmetric information alone is not enough. The wedge comes from the combination of hidden outcomes, limited liability/nonnegative entrepreneur wealth, and costly verification.

4.7 Interest-rate-taking behavior

The paper compares the optimal contract to a borrower who treats the interest rate r as exogenous and chooses investment as if he can borrow any amount at that rate:

$$I \in \arg \max_{\hat{I} \geq 0} E \max\{f(s, \hat{I}) - (1 + r)(\hat{I} - W_0), 0\}. \quad (29)$$

The lender must still break even, and the promised repayment equals the debt payment:

$$Eg_\gamma(s, I) = (1 + i)(I - W_0), \quad (30)$$

$$(1 + r)(I - W_0) = f(\gamma, I) = R_1. \quad (31)$$

The paper shows that the bankruptcy cutoff is higher under interest-rate-taking behavior than under the optimal contract, under the stated conditions. Thus bankruptcy is more likely.

Important distinction. The investment level itself cannot be ranked unambiguously. A rate-taking borrower may overborrow relative to the optimal contract for a given rate, but the equilibrium rate and bankruptcy cutoff also adjust.

4.8 Risk aversion

If the entrepreneur is risk averse and the investor is risk neutral, the strict standard-debt form changes in bankruptcy states. The lender still receives a fixed payment in nonbankruptcy states, but in bankruptcy the entrepreneur may receive insurance instead of losing all residual wealth.

In the risk-averse extension, an optimal contract can involve:

$$B(W_1 - D) = 0 \tag{32}$$

for some constant D , meaning the entrepreneur receives the same wealth in verified states.

What survives. The MEP result continues. The threshold-verification logic continues. The bounds on investment and the underinvestment results continue because they rely on the same conflict between the entrepreneur's preferred investment and the lender's break-even condition.

What changes. Maximum recovery in bankruptcy no longer generally holds, because full extraction in bad verified states exposes the risk-averse entrepreneur to too much consumption risk.

4.9 Liquidity and monotonicity

The paper's final result concerns how investment changes with borrower net wealth W_0 . A simple intuition says more wealth relaxes credit constraints and should raise investment. Gale and Hellwig show this need not hold.

With $c_1 = 0$ and $c_0 > 0$, the first best can arise when W_0 is very high, because the borrower can finance the project safely. But the first best can also arise when W_0 is extremely low and bankruptcy occurs in all states, because the repayment schedule no longer creates a marginal bankruptcy-region distortion. For intermediate wealth, underinvestment can occur.

Economic intuition. Wealth affects not only the amount of outside financing but also the repayment threshold and bankruptcy probability. These channels can offset each other, so the investment-wealth relation may be U-shaped rather than increasing.

5 Identification and interpretation

5.1 What the paper identifies

This is a theoretical identification exercise, not an empirical one. The paper identifies a mechanism: costly verification plus borrower private information implies a debt-like optimal contract and can generate underinvestment.

5.2 Why the costly-state-verification friction matters

Without costly verification, the lender could observe the state and write a complete-contingent contract. Without hidden information, the borrower could not misreport. Without limited liability or nonnegative entrepreneur wealth, the borrower could absorb losses directly. The debt contract emerges because all three features interact.

5.3 Debt as a low-information contract

Debt minimizes the number of states in which the lender needs information. In high-output states, the lender only needs to know that the fixed repayment is made. In low-output states, verification is used to determine how much can be recovered.

5.4 Bankruptcy is costly but useful

Bankruptcy destroys resources, but it also produces information and enforces repayment. The optimal contract uses bankruptcy sparingly: only in states in which the borrower cannot make the fixed payment.

5.5 Meaning of credit rationing

Credit rationing in this paper means the efficient contract chooses an investment level below the complete-information first best. It is a rationing of loan size and project scale, not necessarily a rationing of borrowers out of the market.

5.6 Why diminishing returns are central

Diminishing returns make the marginal product of investment fall as scale rises. Near the first best, a small reduction in investment can reduce the probability and cost of bankruptcy enough to dominate the lost output. This is why the underinvestment result is closely connected to diminishing returns.

5.7 Relationship to Stiglitz-Weiss and Diamond

The model differs from Stiglitz and Weiss because asymmetric information concerns realized revenue, not project choice, and investment is divisible. It differs from Diamond's delegated-monitoring setting because bankruptcy costs here are exogenous and can make the second best inefficient.

5.8 Limitations of the model

The model is one-period, abstracts from repeated lending relationships, and assumes the investment decision itself is contractible. It does not model borrower effort, ex ante adverse selection, renegotiation, or endogenous liquidation costs. These simplifications make the debt result clean but leave richer dynamic questions for later work.

6 Tables and figures

6.1 Title/abstract and model assumptions

Read:

- The title and abstract state the main theorem: the optimal incentive-compatible contract is standard debt.
- The abstract also previews the underinvestment result: second-best investment never exceeds first best and is strictly below first best when costly bankruptcy occurs with positive probability.
- The assumptions define the production function, asymmetric information, and observation costs.

Economic message: The entire paper is built around one friction: the borrower privately observes the state, while the lender can learn it only through costly verification.

6.2 Incentive compatibility and standard debt

Read:

- Proposition 1 shows that all nonobserved states must carry the same lender payment.
- The formal contract problem maximizes entrepreneur payoff subject to investor break-even, feasibility, limited liability, and incentive compatibility.
- The standard debt contract is defined by fixed repayment, bankruptcy exactly when revenue is insufficient, and maximum recovery in bankruptcy.

Economic message: Debt is a contract that keeps the lender uninformed in good states and pays for information only in bad states.

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Incentive-Compatible Debt Contracts: The One-Period Problem

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In a simple model of borrowing and lending with asymmetric information we show that the optimal, incentive-compatible debt contract is the standard debt contract. The second-best level of investment never exceeds the first-best and is strictly less when there is a positive probability of costly bankruptcy. We also compare the second-best with the results of interest-rate-taking behaviour and consider the effects of risk aversion. Finally we provide conditions under which increasing the borrower's initial net wealth must reduce total investment in the venture.

1. INTRODUCTION

A great deal of effort has recently gone into the study of *implicit contract models* (ICM)

Extracted display: title and abstract of the paper.

entrepreneur and she makes a profit at the going rate of interest.

Each entrepreneur is the owner-manager of a single firm. We can think of investment as an input to a production process which yields a random revenue, either because of technological uncertainty or because of uncertainty about the price at which the output will be sold. The production process involves two dates, indexed 0 and 1. At the first date, the level of investment is chosen; at the second, the revenue is observed. At date 0 the entrepreneur's *net wealth* (ignoring the value of the firm as a going concern) is $W_0 = A_0 - R_0$, where $A_0 \geq 0$ is the entrepreneur's liquid assets and $R_0 \geq 0$ the firm's initial indebtedness. The entrepreneur is assumed to be *risk-neutral*; he maximizes the expected value of his "wealth" at date 1. The returns to the risky venture are described by a revenue function f . An investment of I units at date 0 produces a revenue of $f(s, I)$ units in state s at date 1.

Assumption 1. $f: \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice continuously differentiable⁴ on the interior of $\mathbb{R}_+ \times \mathbb{R}_+$, and continuous at the boundary. For any $(s, I) \gg 0$:

- (i) $f(0, I) = f(s, 0) = 0$;
- (ii) $D_2 f(s, I) > 0$, $D_{22} f(s, I) < 0$ and $D_{12} f(s, I) > 0$.

The probability distribution of states⁵ is represented by a cumulative distribution function H with support in $\mathbb{R}_+$.

A crucial assumption is that agents have asymmetric information. At date 0, neither the investor nor the entrepreneur can observe the state. At date 1, the entrepreneur observes the state free of charge. The investor can observe the state only at some cost. In fact, costs may be imposed on the entrepreneur as well as the investor. This makes sense in terms of our interpretation of the event of observing the state as "bankruptcy". In that case there may be some penalty imposed on the debtor for going bankrupt. Let

Extracted display: model primitives and assumptions.

6.3 Optimality of standard debt and reduced problem

Read:

- Proposition 3 shows any optimal contract is weakly dominated by a standard debt contract with maximum equity participation.
- Proposition 4 gives conditions under which the optimal contract itself is a standard debt contract.
- The bankruptcy cutoff representation replaces the full contract by investment I and a cutoff γ .
- The reduced problem separates borrower surplus net of bankruptcy costs from lender recovery.

Economic message: The complicated contract-design problem collapses into choosing a project scale and a bankruptcy threshold.

Therefore the contract is *incentive-compatible* if and only if

$$\begin{aligned} &\text{for any states } s \text{ and } \hat{s} \text{ such that } B(s) = 0 \text{ either} \\ &W(s, \hat{s}) < 0 \text{ or } W(s, \hat{s}) \leq W_1(\hat{s}) - c_0 B(\hat{s}). \end{aligned} \quad (1)$$

Proposition 1. A contract (I, C_0, C_1, W_1, B) is *incentive-compatible* if and only if

- (a) *there exists a constant R_1 such that $C_1(s) = R_1$ whenever $B(s) = 0$ and*
- (b) *for any states s and $\hat{s}$ such that $B(s) = 0$, $B(\hat{s}) = 1$ and $C_1(s) \leq f(\hat{s}, I) + (1+i)(A_0 - C_0)$, $R_1 - C_1(s) \geq c_0 + c_1(s, I)$.*

The proof is immediate from Definition (1). Condition (a) is fairly obvious: if C_1 is not constant across the non-observed states the entrepreneur has an incentive, whenever the value of C_1 is high in a non-observed state, to announce a non-observed state with a lower value. This must be feasible. If the true state is an observed state and it is feasible for the entrepreneur to lie, condition (b) says that the return to the investor must at least absorb the cost of observation. Otherwise the entrepreneur would rather pay a higher value of R_1 to avoid the costs of observation. Thus, although on average observation costs are born by the entrepreneur, at the margin they appear to be born by the investor. In the sequel, we deal only with incentive-compatible contracts.

The *optimal contract problem* is to choose an incentive-compatible contract to maximize the entrepreneur's expected utility subject to the investor's zero-profit condition. Formally, the problem is to choose a contract (I, C_0, C_1, W_1, B) to solve:⁶

$$\max E[W_1 - c_0 B] \quad (2i)$$

$$\text{s.t. } EC_1 \geq (1+i)(I + R_0 - C_0) \quad (2ii)$$

$$C_1 + W_1 \leq f - c_1 B + (1+i)(A_0 - C_0) \quad (2iii)$$

$$0 \leq I, \quad 0 \leq C_0 \leq A_0, \quad 0 \leq W_1 \quad (2iv)$$

$$(I, C_0, C_1, W_1, B) \text{ is incentive-compatible.} \quad (2v)$$

(2ii) is the zero-profit condition in inequality form; (2iii) is a feasibility condition; (2iv) is self-explanatory but note that $0 \leq W_1$ is crucial if the problem is to be non-trivial; and (2v) is the *incentive-compatibility condition*.

Extracted display: incentive compatibility and the optimal contract problem.

Condition (3ii) corresponds to the old condition $W_1 \geq 0$. Note that we are entitled to drop the reference to W_1 in (3v) since it does not appear in the conclusion of Proposition 1.

In order to characterize the form of the optimal contract we need two important concepts. The first is *maximum equity participation* (MEP). A contract (I, C_0, C_1, B) exhibits MEP if and only if $C_0 = A_0$. The second is the *standard debt contract* (SDC). A debt contract in the usual sense has three essential features. When the debtor is solvent a debt contract involves a fixed payment to the creditor. The debtor is declared bankrupt if and only if he cannot make this fixed payment and, in the event of bankruptcy, the creditor recovers as much as he can. A contract (I, C_0, C_1, B) is said to be a SDC if and only if

$$(\text{fixed repayment}) \text{ for some } R_1, (1-B)(C_1 - R_1) = 0 \quad (4i)$$

$$(\text{bankruptcy decision}) B = 1 \Leftrightarrow f < R_1 \quad (4ii)$$

$$(\text{maximum recovery}) BC_1 = B(f - c_1). \quad (4iii)$$

Here we have identified the event $B=1$ with bankruptcy and interpret $f - c_1$ as the *recoverable part of the firm's surplus after bankruptcy*. These two concepts are key to

Extracted display: definition of a standard debt contract.

6.4 Underinvestment and credit rationing

Read:

- Proposition 5 bounds optimal investment between the investor-preferred and entrepreneur-preferred investment levels for a given bankruptcy cutoff.
- Propositions 6 and 7 give sufficient conditions for strict underinvestment.
- The crucial ingredients are positive bankruptcy probability and positive bankruptcy costs.

Economic message: The model generates credit rationing as underinvestment. The contract reduces project scale to limit expected verification losses.

a new contract (l, C_0, C'_1, B') by putting

$$B' = \begin{cases} 0 & \text{if } f \geq R'_1 \\ 1 & \text{if } f < R'_1 \end{cases}$$

and

$$C'_1 = \begin{cases} R'_1 & \text{if } B' = 0 \\ f - c_1 & \text{if } B' = 1. \end{cases}$$

Suppose that $R'_1 = R_1$ where R_1 is the constant value of C_1 when $B = 0$. Whenever $B = B'$, feasibility (3iii) and the construction of C'_1 imply that $C'_1 \geq C_1$. If $B' < B$ incentive-compatibility ensures that $C_1 \geq R_1 \geq C'_1$. Since feasibility implies $B' \leq B$ this proves $C'_1 \geq C_1$ when $R'_1 = R_1$. So we can choose R'_1 to satisfy the zero-profit constraint (3ii). (Choose the smallest value if there is more than one). By construction the new contract is feasible ((3iii) and (3iv)). Incentive-compatibility is satisfied by any SDC with MEP. Then (l, C_0, C'_1, B') is an admissible contract and since, by an earlier argument, $B' \leq B$ it must be optimal.

Proposition 3. Any optimal contract is weakly dominated by a SDC with MEP.

To obtain the converse we need to assume that observation costs are positive and responsive to small changes in the contract.

Proposition 4. Let (l, C_0, C_1, B) be an optimal contract with MEP and let R_1 be the value of C_1 when $B = 0$. For any $\varepsilon > 0$ suppose that

$$\int_{R_1 - \varepsilon \leq f < R_1} (c_0 + c_1) dH > 0.$$

Then (l, C_0, C_1, B) is a SDC.

Extracted display: propositions establishing SDC optimality.

6.5 Interest-rate-taking comparison

Read:

- The interest-rate-taking borrower chooses investment as if funds are available at a fixed rate r .
- The lender's break-even constraint and repayment equation still pin down the feasible contract.
- Proposition 9 implies that interest-rate-taking behavior pushes investment above the lender-preferred level for the associated cutoff.
- Proposition 10 shows that the bankruptcy point is higher under interest-rate-taking behavior than under the optimal contract, under the paper's conditions.

Economic message: Treating the interest rate as exogenous can lead the borrower to ignore how higher promised repayment increases bankruptcy risk.

To see this, note that if $R_1 > f(s, l)$ with probability 1 then $C_0 = A_0$ and the feasibility constraint (3iii) imply that $B = 1$. Then the value of R_1 is immaterial.

In view of (5) and (6) and Assumption 2 there exists a state γ such that $f(\gamma, l) = R_1$. If there is more than one such then let γ be the smallest. Then the contract (l, R_1) can equally well be represented by the ordered pair (l, γ) . Conversely any ordered pair (l, γ) defines a contract if we put $R_1 = f(\gamma, l)$. We call γ the *bankruptcy point* for obvious reasons. It is uniquely determined if $l > 0$. In that case bankruptcy occurs if and only if $s < \gamma$. This interpretation of the bankruptcy point fails only if $l = 0$ in which case $W_0 = 0$, γ is immaterial and there is no possibility of bankruptcy. In either case we can use this new representation to simplify the contract problem even further. For any $\gamma \geq 0$ define functions f_γ and g_γ by putting

$$f_\gamma(s, l) = \begin{cases} f(s, l) & \text{if } s \geq \gamma \\ f(s, l) - c_1(s, l) - c_0 & \text{if } s < \gamma \end{cases} \quad (7)$$

and

$$g_\gamma(s, l) = \begin{cases} f(\gamma, l) & \text{if } s \geq \gamma \\ f(s, l) - c_1(s, l) & \text{if } s < \gamma \end{cases} \quad (8)$$

for any $(s, l) \geq 0$. Then the contract problem (3) can be written compactly as

$$\max E[f_\gamma - (1+i)l] \quad (9i)$$

$$\text{s.t. } Eg_\gamma \geq (1+i)(l - W_0) \quad (9ii)$$

$$(l, \gamma) \geq 0. \quad (9iii)$$

Extracted display: bankruptcy point and reduced optimization problem.

3.1. Comparisons with the first best

The first-best level of investment, I^* , will be achieved if both agents directly observe the state of nature. Then

$$I^* \in \arg \max_{I \geq 0} E\{f(I, s) - (1+i)I\}.$$

For any value of $\gamma \geq 0$ define functions $I^*(\cdot)$ and $I(\cdot)$ by putting

$$I^*(\gamma) = \arg \max_{I \geq 0} E\{f_\gamma(s, I) - (1+i)I\} \quad (10)$$

$$I(\gamma) = \arg \max_{I \geq 0} E\{g_\gamma(s, I) - (1+i)I\}. \quad (11)$$

From Assumptions 1 to 3, $D_2 g_\gamma \leq D_2 f_\gamma \leq D_2 f$ and $I^* < \infty$ so the maximands are bounded above. Since they are also continuous a maximum exists in each case. The maximum is unique because f is strictly concave and c_1 convex in I , so $I^*(\cdot)$ and $I(\cdot)$ are well-defined.

The functions $I^*(\cdot)$ and $I(\cdot)$ mark the boundaries of the region in which the interests of the entrepreneur and investor conflict. For a given value of γ the investor wants I to move in the direction of $I(\gamma)$. For a given value of γ the entrepreneur wants I to move in the direction of $I^*(\gamma)$. An efficient contract must therefore have I lying between $I(\gamma)$ and $I^*(\gamma)$ as the next proposition shows.

Proposition 5. *If (I, γ) is an optimal contract then $I(\gamma) \leq I \leq I^*(\gamma)$.*

Proof. For any value of γ , $I(\gamma) \leq I^*(\gamma)$ because f_γ and g_γ are concave in I and $D_2 g_\gamma \leq D_2 f_\gamma$. If (I, γ) is optimal and $I > I^*(\gamma)$ a small reduction in I raises both (9i) and (9ii) contradicting optimality. A similar argument shows $I \leq I(\gamma)$. $\parallel$

Proposition 5 is the essential tool for demonstrating the existence of under-investment, as the next result shows.

Proposition 6. *Let $(\hat{I}, \hat{\gamma})$ be an optimal contract and suppose that $D_2 c_1(s, I^*) > 0$ and that $\Pr[s < \hat{\gamma}] > 0$. Then $\hat{I} < I^*$.*

Proof. From the definition of $I^*(\cdot)$ it is obvious that $I^*(0) = I^*$. Since $-c_1$ and f are concave in I , $I^*(\cdot)$ is monotonically non-decreasing. Then $I^*(\hat{\gamma}) = I^*$ implies that $I^*(\gamma) = I^*$ for all $0 \leq \gamma \leq \hat{\gamma}$. This means that $ED_2 f_\gamma(s, I^*) = (1+i)$ for all $0 \leq \gamma \leq \hat{\gamma}$. Then $E\{D_2 c_1(s, I^*) | s < \hat{\gamma}\} = 0$, a contradiction. $\parallel$

The proposition shows that if there is a positive probability of bankruptcy and an increase in investment (at the first-best level) increases the cost of bankruptcy in each state, the second-best optimal investment level is strictly less than the first best. It is obvious that the second best never exceeds the first best since $I^*(\gamma) \leq I^*$ for all $\gamma \geq 0$. The conditions of Proposition 6 are not very strong but they rule out cases where, although the costs of bankruptcy are high, the marginal costs are zero, eg., when $c_1 = 0$. The next proposition is designed to deal with this case. We make assumptions about the distribution of states of nature which are stronger than necessary. The argument can be extended to deal with less-well-behaved measures H but only at the cost of some complications which we seek here to avoid. (See Gale and Hellwig, 1983).

Extracted display: first-best comparison and underinvestment propositions.

6.6 Risk aversion and monotonicity

Read:

- Under risk aversion, the contract still has fixed repayment in nonbankruptcy states, but maximum recovery in bankruptcy need not hold.
- The entrepreneur receives insurance in verified states because the investor is risk neutral.
- The monotonicity section shows that investment need not rise monotonically with borrower wealth.

Economic message: Debt is robust to risk-aversion in spirit, but exact bankruptcy payoffs change. Liquidity comparative statics are subtle because wealth affects both financing needs and the bankruptcy threshold.

this relationship when choosing the “optimal” level of investment. The result of such price-taking or more precisely interest-rate-taking behaviour can be described as a special sort of contract. To simplify the discussion we assume $c_0 = 0$ in what follows. We say that a contract (I, γ) is the result of interest-rate-taking behaviour if three conditions are satisfied. First, the entrepreneur chooses I to maximize his expected utility under the assumption that he can borrow any amount at the interest rate r :

$$I \in \arg \max_{I \geq 0} E \max \{f(s, \hat{I}) - (1+r)(\hat{I} - W), 0\}. \quad (13)$$

Second, the investor breaks even on the contract:

$$Eg_\gamma(s, I) = (1+i)(I - W_0) \quad (14)$$

And finally, the notional rate r is the interest rate actually paid by the entrepreneur when he is not bankrupt:

$$(1+r)(I - W_0) = f(\gamma, I) = R_1. \quad (15)$$

It is intuitively plausible and easy to show that under interest-rate-taking behaviour the level of investment is higher than it would be under utility-taking behaviour with the same notional interest rate. But it is not at all clear whether the actual level of investment will

Extracted display: interest-rate-taking contract conditions.

Proposition 10. Let (l, γ) be an optimal contract and $(\hat{l}, \hat{\gamma})$ the result of price-taking behaviour. Then $\gamma \leq \hat{\gamma}$ if the conditions of Proposition 9 are satisfied by $(\hat{l}, \hat{\gamma})$.

Proof. Since $\hat{l} > l^*(\hat{\gamma})$, $(l^*(\hat{\gamma}), \hat{\gamma})$ is feasible. If $\gamma > \hat{\gamma}$ then

$$\begin{aligned} Ef_{\gamma}(s, l) - (1+i)l &< Ef_{\gamma}(s, l^*(\gamma)) - (1+i)l^*(\gamma) \\ &\leq Ef_{\hat{\gamma}}(s, l^*(\hat{\gamma})) - (1+i)l^*(\hat{\gamma}). \quad \parallel \end{aligned}$$

Remark. The assumption $W_0 > 0$ is relaxed in Gale and Hellwig (1983) where it is shown that $\gamma < \hat{\gamma}$ when the following, special, functional forms are used:

$$f(s, l) = s\phi(l) \quad (16)$$

$$c_0 = \alpha, \quad c_1(s, l) = \beta f(s, l), \quad 0 \leq \beta \leq 1, \quad \alpha \geq 0. \quad (17)$$

The comparison of interest-rate-taking and utility-taking behaviour is ambiguous as far as l is concerned. The nearest we can get to a definite result is the following proposition.

Proposition 11. Suppose the conditions of Proposition 10 are satisfied. Then

- (i) $\hat{l} > l$ if $Eg_{\hat{\gamma}}(s, l) > Eg_{\gamma}(s, l)$
- (ii) $\hat{l} < l$ if $Eg_{\hat{\gamma}}(s, \hat{l}) > Eg_{\gamma}(s, l)$.

Proof. The zero-profit constraint implies

$$Eg_{\gamma}(s, l) - (1+i)l = Eg_{\hat{\gamma}}(s, \hat{l}) - (1+i)\hat{l}$$

so (i) implies

$$E\{g_{\gamma}(s, l) - g_{\gamma}(s, \hat{l}) - (1+i)(l - \hat{l})\} > 0.$$

Concavity implies

$$E\{D_{s, \sigma} (s, \hat{l}) - (1+i)(l - \hat{l})\} > 0$$

Extracted display: comparison of optimal and price-taking behavior.

6.7 Nonmonotonicity of investment in net wealth

Read:

- The paper explains why first-best investment can occur at both high and very low levels of W_0 .
- Proposition 13 states that, under the stated assumptions, the optimal investment level cannot be monotonic in W_0 .
- This is a warning against simple comparative statics in financial-constraint models.

Economic message: Internal wealth changes the whole contract, not just the amount of external finance. As a result, the investment response can be nonmonotone.

complicated under risk-aversion. we do not yet know whether analogues to these propositions exist in the risk-averse case.

5. MONOTONICITY

In Gale (1983a and 1983b) it was shown that liquidity constraints reduce a firm's demand for labour in much the same way as quantity constraints. Furthermore, these constraints give rise to "Keynesian" phenomena just as quantity constraints do. For example, fiscal policy has a multiplier effect on employment and output of the sort predicted by simple Keynesian models. The results in the preceding pages show that effective demand failures occur even when bankruptcy is allowed (in Gale, 1983a and 1983b there is a strict "no bankruptcy" condition) and agents write general, optimal contracts. The underinvestment result ($l < l^*$) can also be interpreted as an underemployment result and generalizes the liquidity-constraint analysis of Gale (1983a and 1983b). An interesting question is whether a reduction in liquidity aggravates the under-investment problem. That is, does a fall in W_0 reduce l ? The answer, perhaps surprisingly, is not necessarily.

A simple example will make this clear. Suppose that $c_1 = 0$ and $c_0 > 0$. We assume that both agents are risk-neutral. If W_0 is sufficiently large then $l = l^*$ (Proposition 10). On the other hand, if W_0 is so small that $\Pr[s \leq \gamma] = 1$ then $l(\gamma) = l^*$. More precisely, when $c_1 = 0$, Eg_{γ} is maximized with respect to γ at $\hat{\gamma} = \sup\{s | H(s) < 1\}$. When

$$W_0 = -(1+i)^{-1}(Ef(s, l^*) - (1+i)l^*)$$

the only contracts satisfying the zero-profit constraint have the form (l^*, γ) with $\gamma \geq \hat{\gamma}$. Thus $l = l^*$ for high and low values of W_0 . However, Proposition 9 gives conditions under which $l < l^*$ for some intermediate values of W_0 . Thus, the relationship between l and W_0 is at best U-shaped.

Extracted display: monotonicity discussion.

dently of the other contracts the investor may be involved in. However, risk-neutrality is not an unreasonable assumption to make in the case of investors since it can be justified as a consequence of risk-pooling. It makes less sense in the case of entrepreneurs and indeed is merely a "simplifying" assumption which ought to be relaxed if possible.

Suppose then that the entrepreneur has a strictly concave Neumann-Morgenstern utility $U(W_i)$ with the usual mathematical properties. The derivation of the constraints in the optimal contract problem (3) is unaffected by the change in the objective function so we can see immediately that the optimal contract problem is:

$$\max EU(W_i) \quad (16i)$$

$$\text{s.t. } EC_1 \leq (1+i)(l + R_0 - C_0); \quad (16ii)$$

$$W_i + C_1 \leq f - Bc_1 + (1+i)(A_0 - C_0); \quad (16iii)$$

$$l \geq 0, \quad 0 \leq C_0 \leq A_0, \quad W_i \geq 0; \quad (16iv)$$

$$\text{for some } R_1, \quad B(R - C_1) \leq Bc_1 \text{ and } (1-B)(C_1 - R) = 0. \quad (16v)$$

We now deal with each of the results in Sections 2 and 3 in turn.

Proposition 2 continues to hold. Every optimal contract in the sense of (16) is weakly dominated by a contract with MEP. The proof is the same as before.

Proposition 3 does not hold. Trivially, $C_1 = R_1$ when the firm is not bankrupt (16v) but in the event of bankruptcy we typically have $W_i > 0$ and hence $C_1 < f - c_1$. For example $W_i > 0$ in almost every state if $U'(0) = \infty$. What we do have still is the condition that

$$B(W_i - D) = 0 \quad \text{for some constant } D. \quad (17)$$

Because the entrepreneur is risk-averse and the investor risk-neutral the entrepreneur gets

Extracted display: risk-averse contract problem.

Proposition 13. Suppose that $c_1 = 0$, $c_0 > 0$ and H has a continuous, strictly positive density on $[0, \hat{\gamma}]$. Then the optimal value of l cannot be monotonic in W_0 .

A complete analysis of the relationship between l and W_0 , which amounts to the complete analysis of the structure of the efficient set of contracts, is extremely difficult. In Gale and Hellwig (1983) we have carried out an exhaustive study of the structure of the efficient set for the class of models characterized by (16) and (17). The example given

Extracted display: Proposition 13 on nonmonotonicity.

7 Short summary

- Gale and Hellwig study a one-period credit contract with hidden project outcomes and costly verification.
- The entrepreneur observes the state for free; the lender observes it only by paying a cost.
- The optimal contract is a standard debt contract under the baseline assumptions.
- Nonbankruptcy states involve a fixed repayment because incentive compatibility requires all nonverified states to look the same to the lender.
- Bankruptcy occurs exactly when the borrower cannot make the fixed repayment.
- In bankruptcy, the lender recovers as much as possible in the risk-neutral baseline.
- The contract can be summarized by investment I and a bankruptcy cutoff γ .
- The second-best investment level never exceeds the first-best level.
- With positive probability of costly bankruptcy, the second-best investment level is strictly below the first-best level.
- Credit rationing means the optimal loan and project scale are too small relative to the complete-information benchmark.
- Interest-rate-taking behavior generally implies a higher bankruptcy cutoff than the optimal contract.
- With entrepreneur risk aversion, the exact debt form changes in bankruptcy states, but the underinvestment logic survives.
- The relation between borrower net wealth and investment need not be monotonic.

8 Conclusion

8.1 Core conclusion

The paper shows that the standard debt contract is not merely a legal convention. It can be the optimal incentive-compatible arrangement when borrowers privately observe output and lenders must pay to verify it.

8.2 Economic intuition

Debt economizes on information. In good states, the lender receives a fixed repayment without inspecting the firm. In bad states, verification is worth paying for because it determines recoverable value and prevents the borrower from hiding output.

8.3 Incremental contribution revisited

The paper's major contribution is to connect three ideas in one formal model: endogenous debt, costly bankruptcy, and credit rationing. The same friction that makes debt optimal also distorts

investment, because costly verification makes some positive-NPV investment unattractive in the second best.

8.4 Implications

The model explains why financially constrained firms may invest less than first-best even in competitive capital markets. It also explains why bankruptcy law and verification technology can affect real investment, not just repayment distribution.

8.5 Limitations

The setting is one-period and highly stylized. It abstracts from repeated lending, renegotiation, borrower effort, project choice, and endogenous monitoring technology. The clean result should be read as a benchmark foundation, not a complete theory of all debt contracts.

8.6 Takeaway

The enduring takeaway is that debt and bankruptcy arise naturally when verification is costly. Debt minimizes unnecessary information production, but costly verification creates a real investment wedge, so financial contracting affects the scale of economic activity.